	update(&s);
	printf("%d", s->marks);
	}
	Output of the program?
	A. 50
	B. 60
	C. Garbage value
	D. Compilation error

SET D

Q. No	Question
1	Given a structure in C, how would you dynamically allocate memory for an array of 10 such structures? A. struct myStruct arr[10]; B. struct myStruct* arr = (struct myStruct*)malloc(10 * sizeof(struct myStruct)); C. struct myStruct* arr = malloc(10); D. struct myStruct arr = malloc(10 * sizeof(struct myStruct));
2	Which of the following is true about an Abstract Data Type (ADT)? A. It is a concrete implementation of data structure. B. It specifies the logical behavior of a data structure without specifying the implementation. C. It can only be implemented using arrays. D. It does not support any operations.

3	What is the output of the following code?
	void main()
	{int *p = NULL;
	p = realloc(p, sizeof(int) * 2);
	printf("%d %d", p[0], p[1]);
	p[0] = 1;
	p[1] = 2;}
	
	A. 1 2
	B. 0 0
	C. Segmentation fault
	D. Garbage

4	struct Node {
	int data;
	struct Node next;
	struct Node *next;
	};
	Based on the code above, which of the following statements are correct?
	
	A. The line struct Node next; is valid in C and creates a new Node inside Node.
	B. The line struct Node next; will cause a compilation error due to incomplete type.
	C. The line struct Node *next; is valid and defines a pointer to another Node.
	D. A structure can contain a pointer to itself but not a direct instance of itself.

5	struct Student {
	int id;
	float marks;
	};
	struct Student s[3];
	struct Student *ptr = s;
	Which of the following access the marks of the second student?
	
	A. ptr->marks
	B. (*ptr+1).marks
	C. (ptr + 1)->marks
	D. ptr[2].marks

6	struct Point {
	int x, y;
	};
	
	struct Point pt = {.y = 10, .x = 5};
	What will pt.x and pt.y be?
	
	A. 0, 10
	B. 5, 10
	C. Compilation error
	D. 10, 5

7	What is the time complexity of the following loop? for (int i = 1; i <= n; i *= 2) for (int j = 1; j <= n; j++) printf("*"); A. O(n log n) B. O(n²) C. O(log n) D. O(n)
8	f(n)=6n^2+4n Which of the following statements is true? A. f(n)=O(n^3) but not Θ(n^2) B. f(n)=O(nlogn) and also Θ(nlogn) C. f(n)=O(n^2) but not Θ(n^2) D. f(n)=O(n^2) and also Θ(n^2)
9	10. Given the asymptotic complexity of 4 functions f1, f2, f3, f4 as follows: f1(n) = 2^n f2(n) = n^(3/2) f3(n) = n log n f4(n) = n log log n Arrange them in increasing order of complexity A. f2, f3, f4 , f1 B. f1, f4, f3 , f2 C. f4, f3, f2 , f1 D. f3, f1, f4 , f2
10	What is the value of *(*arr + 1) in this 2D array? int arr[2][3] = {{10, 20, 30}, {40, 50, 60}}; A. 10 B. 20

	C. 40 **D. 50**